

Project
On
Fake News & Deepfake Detector

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Abstract

The rapid spread of fake news and deepfake media on digital platforms has created serious challenges in verifying information authenticity. Traditional fact-checking methods are slow and unable to handle the large volume of online content. Existing AI systems often focus only on text or images, resulting in limited detection accuracy.

This project proposes a Multimodal Fake News and Deepfake Detection System that combines a BERT-based fake news classifier with a CNN-based deepfake image detector. The BERT model analyzes textual content, while the CNN model detects manipulated visual features in images.

A fusion-based ensemble mechanism combines outputs from both models to generate a final REAL or FAKE prediction with confidence scores.

The proposed system improves detection accuracy, supports real-time verification, and provides a reliable solution for identifying misinformation across digital platforms.

Introduction

Background

The growth of social media platforms has accelerated the spread of misinformation. Fake news articles and manipulated images can influence public opinion, cause panic, and create social instability. Traditional verification methods depend on human fact-checkers, which cannot scale to the massive volume of content shared every minute.

Recent AI-based solutions focus on text classification or deepfake image detection independently. However, misinformation often combines **text and images**, requiring multimodal analysis. Furthermore, many systems lack explainability and real-time performance.

Problem Definition & Research Gap

Existing fake news detection systems suffer from the following limitations:

1. Single-modality detection (text-only or image-only)
2. Poor generalization across datasets
3. Lack of multilingual support
4. No explainable decision-making
5. Slow processing for real-time use

There is a need for an intelligent system that can:

- Analyze both text and images
- Support multilingual fake news detection
- Provide explainable predictions
- Operate in real-time environments
- Improve detection accuracy using model fusion

Proposed Solution

This project proposes a **Multimodal Transformer-Based Fake News & Deepfake Detector**. The system combines:

This project proposes a Multimodal Fake News and Deepfake Detection Ensemble System. The system combines:

BERT for fake news text classification

- CNN-based deepfake image detection
- Ensemble fusion mechanism for combining predictions
- Softmax and sigmoid confidence scoring for final decision-making

The BERT model analyzes news articles using transformer-based contextual understanding, while the CNN model extracts visual features from images to identify manipulated content. The outputs from both models are combined using weighted score fusion to generate a final REAL or FAKE prediction along with confidence scores.

The proposed system improves reliability, enhances multimodal misinformation detection, and provides a scalable AI-based solution for trustworthy information verification across digital platforms.

Key Features

- **Multi-modal analysis:** Simultaneously processes text, images.
- **Ensemble architecture:** Combines multiple transformer models for improved accuracy
- **Real-time processing:** Provides rapid classification suitable for live content moderation
- **Explainable outputs:** Generates confidence scores and highlights suspicious elements
- **Cross-platform compatibility:** Works with content from various social media formats
- **Continuous learning:** Model can be fine-tuned as new manipulation techniques emerge

Problem Statement

The current landscape presents several challenges:

1. Deepfake generation technology is becoming increasingly accessible
2. Fake news spreads faster than corrections on social media
3. Human detection accuracy for sophisticated deepfakes is below 50%

Objectives

To address the identified research gaps, this project is guided by the following core objectives:

1. **Multilingual Detection**-To detect fake news in multiple languages using transformer models.
2. **Image Manipulation Detection**-To identify manipulated images using CNN-based Vision Transformers.
3. **Multimodal Analysis**-To analyze both text and images simultaneously.
4. **Weighted Fusion**-To combine text and image predictions effectively.
5. **Confidence Scoring**-To generate probability-based authenticity scores.
6. **Explainable AI**-To provide interpretable detection results.
7. **Real-Time Processing**-To enable fast content verification.
8. **Better Generalization**-To improve performance across datasets.
9. **Reduced Manual Effort**-To automate fact-checking process.
10. **Scalable Deployment**-To support integration with online platforms.

Scope

This project can be used in the following areas:

- News organizations to help automatically check whether news is real or fake
- Social media platforms to control and filter false or misleading content
- Research institutions to study how misinformation spreads
- Government agencies to monitor fake information and online threats

Current Limitations

- The system currently works mainly for English content (support for other languages will be added later)
- Video detection works only when a human face is present
- It needs GPU support for fast and real-time processing

Future Extensions

In the future, this system can be extended with:

- Database integration to store results and training data
- API support so other platforms can connect with the system
- Browser extension to help users check content directly while browsing
- Mobile application for detecting fake content on smartphones .

System Requirements

Hardware Requirements

- RAM: Minimum 4GB (8GB recommended)
- Processor: Intel i3 / Ryzen 3 or higher
- Storage: Minimum 10GB free space
- GPU (Optional): NVIDIA GPU for faster model training and inference

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python
- Development Environment: Jupyter Notebook / Google Colab /VS-Code

Deep Learning Frameworks

- PyTorch
- TensorFlow

Libraries Used

- Hugging Face Transformers
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

Models Used

- BERT for fake news detection
- CNN (Convolutional Neural Network) for deepfake image detection
- Ensemble fusion mechanism for multimodal prediction

Methodology

1. Data Collection

- Collected fake and real news datasets
- Collected real and fake image datasets for deepfake detection

2. Text Preprocessing

- Combined news title and article text
- Removed unnecessary data
- Prepared textual input for transformer processing

3. Tokenization

- Used the tokenizer of BERT
- Applied padding and truncation
- Generated input IDs and attention masks

4. Image Preprocessing

- Resized images to 128×128 pixels
- Normalized image pixel values
- Converted images into NumPy arrays

5. Fake News Detection

- Used BERT for contextual text understanding
- Classified news articles as REAL or FAKE
- Generated probability scores using softmax

6. Deepfake Detection

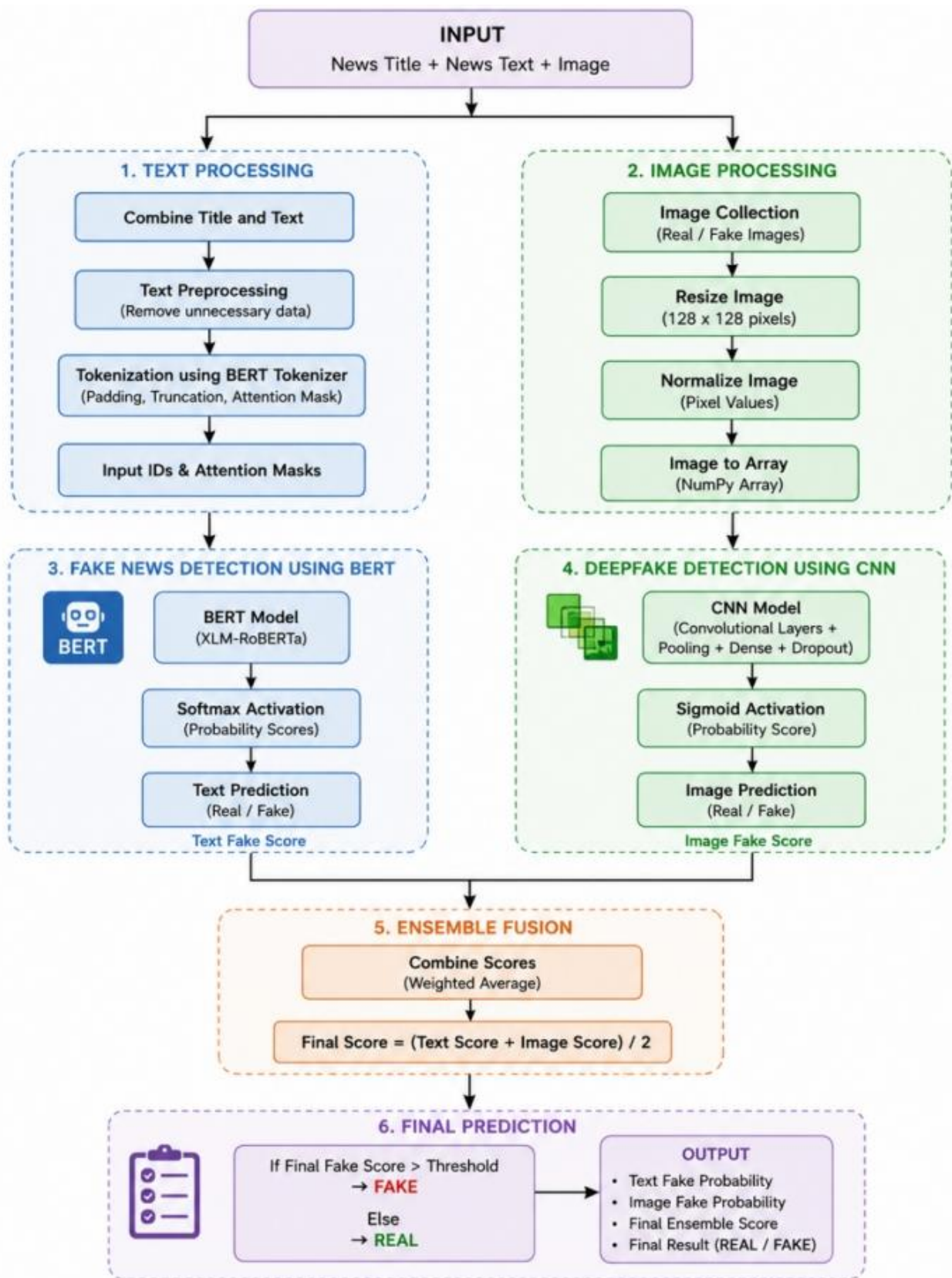
- Used CNN architecture for image classification
- Applied convolution, pooling, flatten, dense, and dropout layers
- Generated fake probability scores using sigmoid activation

7. Ensemble Fusion

- Combined outputs from BERT and CNN models
- Applied weighted score averaging for final prediction

8. Final Prediction

- Generated final REAL or FAKE result
- Displayed confidence scores for both text and image analysis
- Improved overall detection accuracy using multimodal fusion



Flowchart

Implementation(STEPS)

The system consists of the following modules:

1. Data Preprocessing Module

- Collects fake and real news datasets along with real and fake image datasets
- Cleans textual data and combines title with article content
- Resizes and normalizes images for CNN processing
- Converts text and images into model-compatible formats

2. Text Detection Module

- Uses BERT for fake news classification
- Applies tokenization, padding, truncation, and attention masks
- Extracts contextual features from textual content
- Generates REAL or FAKE prediction with confidence score

3. Image Detection Module

- Uses CNN architecture for deepfake image detection
- Applies convolution, pooling, flatten, dense, and dropout layers
- Extracts facial and visual manipulation features
- Generates fake probability score using sigmoid activation

4. Fusion Module

- Combines outputs from text and image models
- Uses weighted average ensemble fusion mechanism
- Produces final multimodal REAL or FAKE prediction

5. Output Module

- Displays:
 - Text fake score
 - Image fake score
 - Final confidence score
 - Final REAL or FAKE result
- Shows analyzed image with prediction label

Timeline Breakdown

Weeks 1-2: Define project scope, gather training datasets from fact-checking organizations and deepfake research repositories

Weeks 3-4: Design transformer architectures, establish training pipelines, set up GPU infrastructure

Weeks 4-5: Implement and train text classification models, evaluate on benchmark datasets

Weeks 5-6: Develop video analysis pipeline, integrate face detection and temporal analysis

Weeks 6-7: Combine models into ensemble system, tune voting weights, optimize inference speed

Weeks 7-8: Comprehensive testing, documentation, deployment to demo environment.

Conclusion

The proposed Fake News & Deepfake Detection System successfully demonstrates an effective **multimodal approach** for identifying misinformation by combining both textual and visual analysis. Unlike traditional single-modality systems, this framework integrates a **BERT-based text classification model** and a **CNN-based image classification model**, improving detection reliability and robustness.

The system applies a **weighted fusion strategy** to combine predictions from both modalities, enabling more accurate final decisions. Additionally, the integration of deep learning models allows the system to handle complex and real-world data patterns effectively.

Although explainability techniques like SHAP and LIME are part of the proposed architecture, the system can be further enhanced by deeper interpretability and real-time deployment optimization. Overall, the project provides a scalable, automated, and intelligent solution for combating misinformation in digital media, contributing to safer and more trustworthy information ecosystems.

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